

CS & IT ENGINEERING



'C' Programming

Control Statements

Lecture No.- 02



By- Satya sir

Recap of Previous Lecture



- Control Statements
 - Classification
 - Conditional Control Statements
 - if
- scanf()



Topics to be Covered



- Nested if
- if-else
- Nested if-else
- Switch - Case



Topic : Control Statements - 2



i j k
 $\boxed{7/3/3}$ $\boxed{7/2/2}$ $\boxed{0/1/1}$

Home work

int i=2, j=3, k=0;

~~0~~

o/p:

Decision Making Expression

Non-zero

if(i++, j--, --k)

$\Rightarrow \text{if}(2, 3, \underline{-1}) \cong \text{if}(\underline{-1})$ True

printf("Hello\n");

if(i, j, k)

$\text{if}(3, 2, \underline{-1}) \cong \text{if}(\underline{-1})$ True

printf("Hai\n");

if(--i, ++j, k++)

$\Rightarrow \text{if}(2, 3, \underline{-1}) \cong \text{if}(\underline{-1})$ True

printf("Hey\n");

i++;

--j;

k++;

$\rightarrow \text{if}(i, j, k)$

printf("Bye");

$\text{if}(3, 2, \underline{1}) \Rightarrow \text{if}(\underline{1})$ True

o/p:

Hello

Hai

Hey

Bye



Nested-if

- if inside another if, that inside another if and so on - - -

Syntax:

```
if (Expression 1)
{
    if (Expression 2)
    {
        if (Expression 3)
        {
            ...
        }
    }
}
```

Ex:

```
int x=7, y=2, z=5;
if (x > y) True
{
    if (x > z) True
    {
        if (y < z) True
        {
            printf("%d", x);
        }
    }
}
```

o/p: 7



Topic : Control Statements - 2



Ex:

```
if (Exp1)
{
    Statement A;
    if (Exp2)
    {
        Statement B;
        if (Exp3)
        {
            Statement C;
        }
        Statement D;
    }
    Statement E;
}
```

<u>Exp1</u>	<u>Exp2</u>	<u>Exp3</u>	<u>Statements executed</u>
F	F	F	—
F	T	F	—
F	F	T	—
F	T	T	—
T	F	F	A, E
T	F	T	A, E
T	T	F	A, B, D, E
T	T	T	A, B, C, D, E



Topic : Control Statements - 2

EX:

⁻¹~~0~~
⁻¹~~0~~
¹~~0~~
int i = ~~0~~, j = ~~-1~~, k = ~~1~~;

if(++i) if(1) True

{

printf("Hai");

if(j++) if(-1) True

{ printf("Hey");

if(--k) if(0) False

{ printf("Hello"); X

}
if(i--, --j, k++)
if(0, -1, 0) False

printf("Bye"); X
}

if(j, k, i--) if(-1, 1, 0) False
printf("Hi"); X

printf("%d %d %d", i, j, k);

}

o/p: Hai Hey -1 -1 1



Topic : Control Statements - 2



/* Find and Print biggest of 2 Numbers */

```
int a, b;
```

```
scanf("%d %d", &a, &b);
```

Wrong
Logic

```
if (a > b)
    printf("a is big\n");
printf("b is big");
```

→ It always
execute

o/p: i/p: 5 < 7

o/p: b is big

i/p: 7 < 5

o/p: a is big
b is big

Wrong o/p

if (a > b) True / False

→ printf("a is big"); ✓

if (b > a) False / True

→ printf("b is big");

if-else

if (a > b)

printf("a is big");

else

printf("b is big");

i/p: 5 < 7

o/p: b is big

i/p: 7 < 5

o/p: a is big

Correct o/p always



Topic : Control Statements - 2



```

if(a > b)      ==> Exp 1
    printf("a is big");
else
    printf("b is big");
    
```

True, Exp 2
false, Exp 3

Replace

```

int a = 7, b = 3, c;
if(a > b)
{ c = 9;
}
else
{ c = 10;
}
    
```

$\Rightarrow a > b ? c = 9 : c = 10 ;$
Error.

$a > b ? \text{printf("a is big")} : \text{printf("b is big")};$
 $a > b ? c = 9 : c = 10$

NOTE:

Each if-else Pair can be replaced with $? :$ (Except Assignment statements)

Ex:

```

if(a > b)
{ printf("Hai");
  printf("Hello");
}
    
```

```

else
{ printf("Hey");
  printf("Bye");
}
    
```

$\Rightarrow a > b ? \text{Pf("Hai"), Pf("Hello")} : \text{Pf("Hey"), Pf("Bye")};$

$a = 7, b = 5$ o/p Hai, Hello Bye

$a = 5, b = 7$ o/p: Hey Bye

Independent Stmt



Topic : Control Statements - 2



```
int x=3, y=7;
```

```
if(x > y)
```

```
    printf("Hello"); ✓
```

```
    printf("Hai"); ← misplaced
```

```
else
```

```
    printf("Bye");
```

```
// Error, ***  
else must be immediately after if.
```

```
if(x > y)  
{  
    pf("Hello");  
    pf("Hai");  
} ✓  
else  
    pf("Bye");
```

Works
Fine



Nested if-else

```
if (a > b)
{
    if (a > c)
        Pf("a is big");
    else
        Pf("c is big");
}
else
{
    if (b > c)
        Pf("b is big");
    else
        Pf("c is big");
}
```

if-else-if ladder

```
if (Exp1)
{
    Stmt A;
    ;
}
else if (Exp2)
{
    Stmt ;
    ;
}
else if (Exp3)
{
    Stmt ;
    ;
}
else
{
    ;
}
```

```
if (rollno == 1)
    Pf("Rama");
else if (rollno == 2)
    Pf("Rehman");
else if (rollno == 3)
    Pf("Robert");
else if (rollno == 4)
    Pf("Ramya");
else
    Pf("Roy");
```




2 mins Summary



- Nested if
- if-else
- Nested if-else
- if-else-if ladder

To be contd . . .





THANK - YOU